

5(a)(i)	product of density and speed	M1
	speed of sound in the medium	A1
5(a)(ii)	$Z_B = 1.8 \times 10^3 \times 4.1 \times 10^3$ $= 7.4 \times 10^6 \text{ kg m}^2 \text{ s}^{-1}$	A1
5(b)	$\alpha = (1.7 - 1.3)^2 / (1.7 + 1.3)^2 = 0.018$ fraction = 0.98	A1
5(c)(i)	reduction in power/intensity (of wave)	M1
	as the wave passes through the medium	A1
5(c)(ii)	1. ratio = $e^{-\mu x}$	C1
	= 0.90	A1
	2. ratio = 0.62	A1
5(d)	fraction = $0.898 \times 0.617 \times 0.98$ = 0.54	A1